

Sanjay Suresh

ssuresh27@wisc.edu · (608) 733-8816

EDUCATION

University of Wisconsin–Madison

M.S. in Professional Quantum Computing (MSPQC)

Madison, WI

Starting Fall 2026

University of Wisconsin–Madison

B.S. in Computer Science and Mathematics; Physics Certificate

Madison, WI

Aug 2022 – May 2026

- **Select Coursework:** Intro to Quantum Computing, Systems Architecture for Quantum Computing, Quantum Mechanics, Cryptography, Algorithms, Analysis, Linear Algebra & Differential Equations

RESEARCH EXPERIENCE

Argonne National Laboratory

Quantum Computing Research Assistant — Distributed Quantum Computing

Lemont, IL

2025 – Present

- Advised by Dr. Rajkumar Kettimuthu. SeQUeNCe: **2026 R&D 100 Finalist**.
- Designed and built a distributed quantum computing module with a pluggable partitioner → scheduler → compiler stack, mapping monolithic circuits onto multi-node quantum networks under per-node memory constraints. Supports telegate, teledata, multi-hop entanglement swapping, and purification.
- Implemented multiple placement strategies (topology-aware capacity/distance allocation, QAP, pytket-dqc) and scheduling strategies (ALAP, packed, resource-aware), including a time-sliced FGP-rOEE hybrid that re-partitions qubits per slice via relaxed OEE local search.
- Validated by running distributed Grover's search up to **11 qubits** and reproducing the **71% two-qubit Grover accuracy** from a published Nature DQC experiment via density matrix formalism. Presented at the 2025 Chicago Quantum Summit.

University of Wisconsin–Madison

Quantum Computing Research Assistant — Benchmarking & Quantum Annealing

Madison, WI

Summer 2023 – 2025

- Advised by Prof. Swamit Tannu. Developed benchmark suites for fault-tolerant quantum hardware; leveraged variational autoencoders to correlate classically simulable metrics with hardware performance.
- Advised by Prof. Krishnan Suresh. Adapted quantum annealing algorithms for constrained optimization with applications to finance portfolio problems; improved linear system solving by **30–40%**. Presented at WCCM Vancouver 2024.
- Taught a workshop on quantum computing for engineers and presented a PyEncode poster at QC 2026.

University of Oxford (Remote Collaboration)

Research Collaborator — Graph State Verification (Dr. Sathyawageeswar Subramanian)

Oxford, UK

Apr 2026 – Present

- Extending graph state verification protocols to more sophisticated Pauli noise models for fault-tolerant measurement-based quantum computing.

PUBLICATIONS

Quantum Science & Technology: K. Suresh, S. Suresh. “PyEncode: An Open-Source Library for Structured Quantum State Preparation.”

JCISE: S. Suresh, K. Suresh. “Computing a Sparse Approximate Inverse on Quantum Annealing Machines.”

Engineering Optimization: S. Suresh, K. Suresh. “Optimal Box Contraction for Solving Linear Systems via Simulated and Quantum Annealing.”

PROJECTS

Studia

Next.js, TypeScript, Convex, Vercel AI SDK, Manim

- Full-stack AI tutoring platform with structured lessons, PDF context extraction via RAG, automated review generation, and real-time Manim animations. **Deployed in CS 639** (Systems Architecture for Quantum Computing, Prof. Swamit Tannu).

LEADERSHIP, TEACHING & AWARDS

- **Atlas Fellowship (2022):** Selected from **7,000+ applicants** worldwide (**114 recipients**) for a \$50K STEM fellowship.
- **Printable Programming (2019 – Present):** Authored *A Visual Approach to Programming for Middle and High School Students* (ISBN: 979-8470081841); taught STEM at the Boys and Girls Club of Dane County.
- **Science Bowl — President & Captain (2020 – 2022):** Ranked **2nd nationally out of 700+ individuals**; founded Summit Scibowl for advanced training; built platforms engaging 2,000+ students.
- **Olympiads & Competitions:** AIME Qualifier; Chemistry Olympiad National Qualifier (2021 & 2022); National Astrophysics Olympiad Qualifier (2021); Earth Science Olympiad Training Camp Qualifier (2022); Wisconsin Math Talent Search Finalist (2022); Mathematics Student of the Year (Middleton HS, 2022).

SKILLS

Languages & Quantum: Python, TypeScript, C, Java, Haskell; Qiskit, PennyLane, D-Wave Ocean, QuTiP, Stim, PyMatching

Web & Tools: Next.js, React, Convex, Vercel AI SDK, Flask, LangChain, LangGraph, Tailwind CSS

OTHER

Carnatic Violin: Professional violinist and accompanist to frontline artists at major festivals across India and the US.